WORKED SOLUTION

STEP II 2015, Question 13

Probability and Statistics • Exponential Distribution Moments

STEP II 2015, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q13

The key idea

The cost is a fixed amount ky plus a random amount that is zero unless the flood overtops the preparations. So everything comes down to the moments of $W = \max(X - y, 0)$, the excess of the flood over the prepared height. The exponential distribution is *memoryless*: given that $X > y$, the excess $X - y$ is again exponential with the same parameter. That turns both the expectation and the variance into one-line calculations.

Solution

Write $p = \mathrm{P}(X > y) = \int_y^\infty \lambda \mathrm{e}^{-\lambda x}\,\mathrm{d}x = \mathrm{e}^{-\lambda y}$, and let

$$W = \begin{cases} X - y & \text{if } X > y, \\ 0 & \text{if } X \le y, \end{cases} \qquad \text{so that} \qquad C = ky + aW.$$

Here $y \ge 0$ is a constant chosen in advance, so ky is not random.

Part (i)

Since $W = 0$ when $X \le y$,

$$\begin{aligned} \mathrm{E}(W) &= \int_y^\infty (x - y)\,\lambda \mathrm{e}^{-\lambda x}\,\mathrm{d}x = \int_0^\infty u\,\lambda \mathrm{e}^{-\lambda(u+y)}\,\mathrm{d}u \qquad (u = x - y) \\ &= \mathrm{e}^{-\lambda y} \int_0^\infty u\,\lambda \mathrm{e}^{-\lambda u}\,\mathrm{d}u = \mathrm{e}^{-\lambda y} \cdot \frac{1}{\lambda}, \end{aligned}$$

using the fact that the mean of an exponential distribution with parameter λ is $\frac{1}{\lambda}$ (or integrate by parts). Therefore

$$\mathrm{E}(C) = ky + a\,\mathrm{E}(W) = ky + \frac{a}{\lambda}\mathrm{e}^{-\lambda y},$$

as required.

Minimising. Let $g(y) = ky + \frac{a}{\lambda}e^{-\lambda y}$ for $y \geq 0$. Then

$$g'(y) = k - ae^{-\lambda y}, \qquad g''(y) = a\lambda e^{-\lambda y} > 0,$$

so g' is strictly increasing, from $g'(0) = k - a$ towards k as $y \to \infty$.

> **Ask yourself** Why can't we just set $g'(y) = 0$ and stop?

Because y is restricted to $y \geq 0$, and the stationary point $e^{-\lambda y} = k/a$ only lies in that range when $k/a \leq 1$. So we split by the value of a/k.

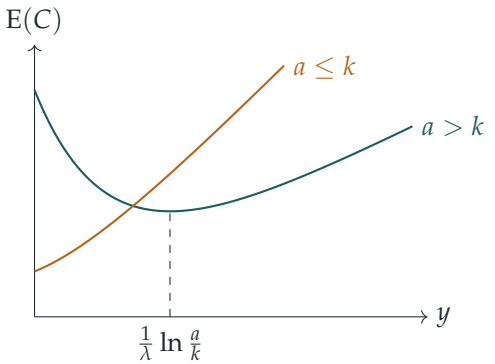
- $a/k > 1$. Then $g'(0) = k - a < 0$, and $g'(y) = 0$ at

$$y = \frac{1}{\lambda}\ln\frac{a}{k} > 0.$$

  Because g' is increasing, g decreases before this point and increases after it, so this is the global minimum on $y \geq 0$. The minimum expected cost is $\frac{k}{\lambda}\left(1 + \ln\frac{a}{k}\right)$.

- $a/k \leq 1$. Then $g'(y) > g'(0) = k - a \geq 0$ for all $y > 0$, so g is strictly increasing on $y \geq 0$ and the minimum is at $y = 0$: make no preparations at all. The expected cost is then $\frac{a}{\lambda}$.

This makes sense: each extra metre of preparation costs k for certain, and saves a only if the flood would have reached it. If $a \leq k$, the saving can never be worth the outlay.



Part (ii)

Adding the constant ky does not change the variance, so $\mathrm{Var}(C) = a^2\,\mathrm{Var}(W)$. For the second moment, with the same substitution $u = x - y$,

$$\mathrm{E}(W^2) = \int_y^\infty (x-y)^2\lambda e^{-\lambda x}\,dx = e^{-\lambda y}\int_0^\infty u^2\lambda e^{-\lambda u}\,du = e^{-\lambda y}\cdot\frac{2}{\lambda^2},$$

since an exponential variable U with parameter λ has $\mathrm{E}(U^2) = \mathrm{Var}(U) + \mathrm{E}(U)^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2}$. Hence, with $p = e^{-\lambda y}$,

$$\mathrm{Var}(C) = a^2\left(\frac{2p}{\lambda^2} - \frac{p^2}{\lambda^2}\right) = \frac{a^2}{\lambda^2}e^{-\lambda y}\left(2 - e^{-\lambda y}\right).$$

The amount spent in advance is ky, which increases with y because $k > 0$; so we must show that $\mathrm{Var}(C)$ decreases as y increases. Differentiating,

$$\frac{\mathrm{d}}{\mathrm{d}y}\left(2\mathrm{e}^{-\lambda y} - \mathrm{e}^{-2\lambda y}\right) = -2\lambda\mathrm{e}^{-\lambda y} + 2\lambda\mathrm{e}^{-2\lambda y} = -2\lambda\mathrm{e}^{-\lambda y}\left(1 - \mathrm{e}^{-\lambda y}\right),$$

which is strictly negative for $y > 0$, since then $0 < \mathrm{e}^{-\lambda y} < 1$. So $\mathrm{Var}(C)$ is strictly decreasing in y on $y \geq 0$: the more spent on preparation, the smaller the variance of the cost, which completes the proof. (Equivalently, $p(2-p) = 1 - (1-p)^2$ increases with p on $[0,1]$, and $p = \mathrm{e}^{-\lambda y}$ decreases with y.)

Checks

- At $y = 0$, $C = aX$, so $\mathrm{E}(C) = a/\lambda$ and $\mathrm{Var}(C) = a^2/\lambda^2$, both matching the formulas.
- As $y \to \infty$ the variance tends to 0: a huge flood defence leaves essentially no uncertainty, only the fixed cost ky.
- With $k = 1$, $a = 3$, $\lambda = 0.5$, $y = 1.2$ the formulas give $\mathrm{E}(C) \approx 4.493$ and $\mathrm{Var}(C) \approx 28.67$; a computer simulation of a million years gave 4.496 and 28.68. For these values the optimal height is $2\ln 3 \approx 2.197$, confirmed by a computer search.

Common slips

- Integrating $a(x-y)\lambda\mathrm{e}^{-\lambda x}$ from 0 instead of from y: there is no flood damage when $X \leq y$.
- Forgetting the boundary $y \geq 0$ and quoting $y = \frac{1}{\lambda}\ln\frac{a}{k}$ even when $a < k$, where it is negative.
- Including ky in the variance; it is a constant and contributes nothing.

Beyond STEP

The quantity $\mathrm{E}\big(\max(X-y, 0)\big)$ is exactly the payoff of a call option with strike y, and the optimum $\mathrm{P}(X > y) = k/a$ is the classic "newsvendor" rule from operations research: choose the level at which the marginal cost of preparing equals the expected marginal saving. The memoryless property behind the neat answers is special to the exponential distribution among continuous distributions. Grimmett and Welsh, *Probability: An Introduction*, covers the exponential distribution and expectations of functions of random variables.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.